Huaiyuan Jing

(929) 426-9451 | Urbana-Champaign, IL | ls.hyjing@gmail.com | github.com/Huaiyuan-Jing | huaiyuan-jing.github.io

Education

University of Illinois Urbana-Champaign Urbana-Champaign, IL
Master of Computer Science *Aug 2026 — Present*

University of Wisconsin-Madison Madison, WI
B.S. Computer Science; B.S. Mathematics *Sep 2022 — May 2026*

- GPA: 3.8/4.0 | US Employment Authorization: Yes
- Relevant Coursework: Algorithms, Operating Systems, Quantum Architecture, Linear Algebra, Probability Theory

Honors & Awards

Gold Medal, ICPC North Central North America Regional (Top 1%) 2024--2025

Silver Medal, ICPC North Central North America Regional 2025--2026

- Ranked elite among university teams, excelling in dynamic programming, advanced data structures, and edge-case handling under extreme time constraints and psychological pressure.

Experience & Research

Research Assistant Feb 2025 — Present
Madison Experimental Mathematics (MXM) Lab *Madison, WI*

- Architected a high-throughput, concurrent toolkit to compute cryptographic hash values for complex graphs and evaluate combinatorial game strategies (Grundy numbers), **successfully evaluating 50 critical data points under extreme $O(2^m)$ time-complexity workloads**.
- Leveraged Rust's Send/Sync traits and ownership model to eliminate data races and deadlocks, designing memory-safe parallel processing streams that maximize multi-core CPU and GPU saturation.
- Optimized graph traversal routines with low-latency lock-free structures, minimizing algorithmic latency under high-complexity, multi-threaded workloads.

First Author Summer 2026
Quantum Computing Algorithm Optimization Research *Madison, WI*

- Proposed an optimized method for single-ancilla block-encoding overhead reduction, successfully reducing the required auxiliary qubit overhead to **exactly 1 ancilla qubit** via polar decomposition.
- Coherently reused block-encoding ancilla to produce a unified **$(1, 1, \varepsilon)$-block encoding** for general matrices, cleanly distributing a total error budget of ε into **two $\frac{\varepsilon}{2}$ branches** to guarantee strict linear error propagation.
- Conducted rigorous query complexity and mathematical error bounds analysis, bounding the polynomial degree for absolute-value approximation to $O\left(\frac{1}{\delta_{\text{abs}}}\log\left(\frac{1}{\eta_{\text{abs}}}\right)\right)$ to prove optimal computational scaling.

Projects

Independent Developer, High-Frequency Trading (HFT) Matching Engine & Order Book (codeberg.org/HYJING/hft_orderbook) Jul 2026

- Designed and implemented an ultra-low latency, fixed-point limit order book (LOB) matching engine from scratch using Modern C++ (C++20).
- Achieved an execution throughput of **38.55 million orders per second (38.5496M/s)** under intensive multivariable cross-order stress tests, benchmarked via Google Benchmark on an 8-core CPU system.
- Optimized memory layout by combining `std::map` for price levels with custom intrusive doubly-linked lists for strict FIFO order queues, minimizing lookup and allocation overhead to $O(1)$ for cancellation and insertion.
- Engineered a vector-backed object-pool memory allocator (`VectorOrderBook`) to completely eliminate runtime OS heap allocations, preventing memory fragmentation and maximizing CPU L1/L2 data cache hit rates.

Independent Developer, Linux Infrastructure & Kernel Optimization Systems Automation

- Maintained a security-critical initramfs module (Dracut-Cryptsetup-Duress) for LUKS-encrypted subsystems, demonstrating deep understanding of the Linux boot process and robust scripting.